

Connor Petri

The Science of Interstellar Questions

1. How does Natalie Batalha find planets?
Using space telescopes and using mass spectroscopy to look for “shadows” of the planets.
2. According to her, what makes a planet habitable?
The presence of liquid water.
3. What is special about planet Kepler 16B?
It orbits a binary star system consisting of a G and M class star.
4. What is planet Kepler 10B made of?
Rock and metal – mostly iron.
5. What makes planet 186F interesting?
It is an Earth-sized planet that sits in the Goldilocks zone of it’s star. It has the conditions for liquid water.
6. What is gravity according to Einstein?
The warping of the fabric of space-time.
7. How does our GPS test this theory?
GPS satellites experience time faster by 38 microseconds per day.
8. What is the twin paradox?
If we sent one twin out on a journey at a relativistically high speed, he would age slower than the twin who stayed on Earth.
9. What is a wormhole?
A particular way in which spacetime can be curved. Theoretically allows information transfer over large distances *apparently* faster than the speed of light, but not actually doing so from it’s frame of reference.
10. What would it take to keep a wormhole open?
An immense amount of gravity and anti-gravity on each side respectively.
11. What does a wormhole look like?
It looks like a crystal ball hovering in space. It is a 3d hole, which is a sphere.

12. What is a blackhole?

When mass is compressed smaller than it's Schwarzschild radius, the gravitational field generated by the mass is so great that not even light can escape, giving the black hole it's name.

13. What is a singularity?

A mass compressed beyond it's Schwarzschild radius is known as a gravitational singularity, which is currently modeled as a one-dimensional point. This model has been challenged as black holes can spin, meaning it has to be more than one-dimensional.

14. What is the event horizon of a blackhole?

When light falls below the Schwarzschild radius, by definition it can no longer ascend above that radius again. This gives the black hole it's color.

15. How does Andrea Ghez study the black hole at the center of our galaxy?

She observed all the stars surrounding it, which allowed her and her team to derive a lot of information by matching the black hole's properties to the motion they observed in the stars.

16. What is the mass of the blackhole at the center of our galaxy?

~4 million solar masses.

17. What is an accretion disk?

A disk of matter that orbits an orbital body.

18. Why does the accretion disk around a black hole look warped?

The path light takes through the universe is warped by the black hole's gravity. This is called gravitational lensing, and happens to all large gravitational bodies.

19. Why would a person falling into a black hole get spaghettified?

The tidal force on their body, or the difference in gravitational force on the feet vs the head, cause the body to be stretched into a thinner and longer string.

20. What is a gravity wave?

When significant changes in a gravitational field are experienced, this sends waves that propagate using the fabric of space-time as a medium. A binary star system is a great example of a system that generates gravitational waves.